Development of an siRNA Database via LLM-Driven Patent Mining

Filipe Kuhlmann

Universidade do Minho, Braga, Portugal
pg42492@uminho.pt

Abstract. Small interfering RNAs (siRNAs) are short, double-stranded RNA molecules capable of silencing specific genes by triggering the degradation of messenger RNA (mRNA) prior to translation. While siRNAs offer immense therapeutic potential, particularly when chemically modified, accurately predicting *in vitro* efficacy remains challenging. Current models often fail to account for chemical modifications and experimental background conditions due to the lack of large datasets integrating these variables.
To address this gap, we propose an automated, Large Language Model (LLM)-driven pipeline to mine rich, unstructured patent literature. To demonstrate its viability, we present a proof-of-concept case study targeting the patent portfolio of Alnylam Pharmaceuticals, the current industry leader. Our goal is to create a pipeline that extracts modified sequences, efficacy metrics, and metadata to generate a robust dataset. Ultimately, we expect our work to establish a siRNA dataset that can be scaled significantly in future work, providing a path to mitigate current dataset biases and facilitate the development of robust deep learning predictive models.

Keywords: Small Interfering RNA (siRNA) · Large Language Models (LLMs) · Patent Mining · Automated Data Extraction

1 Introduction and State of the Art

1.1 Molecular Mechanism of siRNA-Mediated RNA Interference

RNA interference (RNAi) is a fundamental cellular process of gene regulation. This process enables most eukaryotic organisms to silence specific genes by degrading the messenger RNA (mRNA) transcripts without altering the genome [18, 6]. The pathway begins when double-stranded RNA (dsRNA) enters the cytoplasm of a eukaryotic cell and is processed by the host ribonuclease Dicer. Dicer cleaves the dsRNA into small interfering RNAs (siRNAs), which are short double-stranded RNA fragments [22, 28, 35]. These fragments are typically 19 to 23 base pairs long and feature characteristic 2-nucleotide overhangs at the $3'$ ends [22, 35] (Figure 1).

One of these siRNA molecules subsequently binds in the cytoplasm to a protein complex known as the RNA-induced silencing complex (RISC) [17]. Within

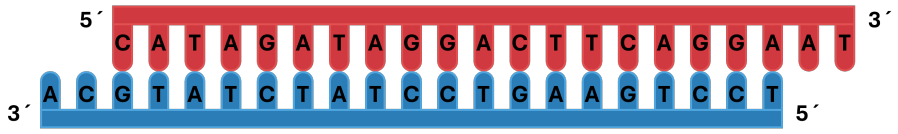

Fig. 1. Structural conformation of a small interfering RNA (siRNA). The antisense (guide) strand is shown in blue and the sense (passenger) strand in red; both strands feature characteristic 2-nucleotide overhangs at the respective 3′ termini.

RISC, the siRNA unwinds; the sense (passenger) strand, which possesses a sequence identical to the target mRNA, is cleaved and discarded. Meanwhile, the antisense (guide) strand, which is complementary to the target mRNA, is retained in the complex [19, 9]. Using this strand, RISC binds to the target mRNA transcript through complementary base pairing. Argonaute 2 (AGO2), an endonuclease belonging to RISC, then cleaves the target mRNA, preventing translation and effectively silencing the gene [19, 32]. Crucially, RISC remains active post-cleavage, allowing the complex with the guide strand to catalyze multiple further rounds of mRNA degradation [11].

1.2 siRNA Therapeutics: Opportunities and Challenges

The discovery in 2001 that synthetic siRNAs can achieve precise, targeted gene silencing in human cells established the foundation for a new class of therapeutics [10]. By leveraging this mechanism, researchers can, in theory, silence virtually any target gene [18, 15]. However, in their unmodified state, siRNA molecules are rapidly degraded by ubiquitous ribonucleases (RNases) in the human body and can even exhibit a half-life of less than 3 minutes [14, 12]. Furthermore, siRNAs are large, negatively charged macromolecules that struggle to cross biological membranes and tissue barriers, such as the blood-brain barrier, to reach target cells [9]. This resistance to membrane permeation presents a substantial delivery challenge. Additionally, the introduction of unmodified siRNA can trigger innate immune responses and cytotoxicity, posing significant safety concerns [1, 12].

To address these inherent limitations, researchers have developed a range of chemical modifications to improve the pharmacokinetic properties and safety profiles of siRNA [12]. For example, at the ribose level, substituting the 2′-hydroxyl (2′-OH) group with alternatives such as 2′-O-methyl (2′-OMe) or 2′-fluoro (2′-F) has proven effective in extending the half-life of molecules, enhancing target binding affinity, and reducing off-target effects, such as the silencing of unintended genes [15, 33, 9, 12]. Beyond chemical modifications, the encapsulation of siRNA within lipid-based delivery systems, such as lipid nanoparticles (LNPs), serves as another major strategy to overcome limitations of unmodified siRNA. LNPs can physically protect siRNA against degradation by RNases and facilitate efficient cellular uptake. Delivery efficiency can be further enhanced by combining LNPs with antibodies or ligands that specifically bind to receptors on the surface of desired target cells [9].

Driven by these advancements in chemical modifications and delivery platforms, significant clinical milestones have been achieved. These include the approval of siRNA-based therapeutics, such as patisiran and givosiran, by the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA) [16, 26]. In recent years, several other siRNA drugs have gained regulatory approval, and numerous candidates are currently undergoing evaluation across various phases of clinical development [9, 33].

Despite recent successes, the design of optimal siRNA sequences remains a significant challenge [25]. Traditional experimental screening entails substantial financial burdens and requires extensive research hours to identify optimal target sites within a given sequence [30, 29, 12]. To mitigate this bottleneck, various *in silico* prediction tools have been developed, ranging from linear regression to artificial neural networks, demonstrating notable predictive success [24].

However, many existing models are trained on datasets composed solely of unmodified siRNA sequences. Consequently, they lack the capacity to account for the complex, state-of-the-art chemical modification patterns currently employed for modern siRNA therapeutics [21, 2]. The predictive accuracy of these models for chemically modified siRNA is often limited, as algorithms trained on non-modified datasets frequently lack predictive power for modified sequences, and vice versa [27]. While models that incorporate chemical modifications have recently emerged [8, 23, 20], most do not consider detailed experimental conditions, from delivery methods to the assays used to measure silencing efficacy. Accounting for these variables is crucial, as siRNA efficacy can be highly assay-dependent and context-specific [5]. For instance, experimental readouts can vary substantially between reporter-based systems, such as luciferase assays, and native mRNA-based assays, such as quantitative reverse transcription polymerase chain reaction (RT-qPCR) [2, 5]. Consequently, it is essential for predictive algorithms to systematically incorporate metadata regarding experimental conditions, as contextual data has been shown to significantly enhance the predictive capacity of a model [3].

To overcome these limitations and accurately predict therapeutic efficacy, it is necessary to develop robust deep learning models that account for both chemical modifications and experimental contexts. However, building these models requires the curation of large-scale, diverse datasets that are notoriously scarce [12]. Fortunately, this critical combination of structural and experimental data is abundantly documented in patent literature, positioning patent mining as an essential strategy to overcome this data bottleneck.

1.3 Patent Mining as a Strategic Resource and Current Limitations

Fundamentally serving as detailed technical records of novel inventions, patents offer a vast and continuously updated repository of siRNA molecules and valuable experimental metadata. They frequently represent the earliest public disclosure of research from various commercial and academic institutions [7]. Since patents enable industries to safeguard intellectual property [4, 7], leading companies at the forefront of siRNA-based therapies, such as Alnylam Pharmaceuticals

[33], extensively document their cutting-edge chemical modification patterns and efficacy screening results within these filings. Despite their value, mining these documents presents a significant challenge, as patents are a mixture of structured and unstructured data [4]. In patent applications, siRNA experimental data is frequently embedded within tables that lack standardization across different applicants and often even across different patents from the same applicant. Key challenges include inconsistent column naming conventions and fundamentally different methods of reporting similar experimental conditions and outcomes.

One recent effort to overcome this barrier is the CMsiRNAdb repository, which catalogs 43,153 experimentally validated, chemically modified siRNA sequences and their corresponding efficacy data derived from 90 international patents [12]. To facilitate data curation from patent literature, the authors developed ModMapper, a custom parsing tool designed to translate the diverse chemical notations found in patents into standardized RNA sequences. This tool is particularly valuable given that chemical modification nomenclatures often vary significantly between different companies and research groups. Despite these advancements, the dataset construction relied on manual workflows, from initial patent retrieval to final quality control. This dependence on manual intervention prevents the platform from scaling across the vast landscape of unmined patent literature. As demonstrated in Figure 2, these 90 patents are a mere fraction of the broader siRNA patent landscape, which inherently limits the scope of the dataset.Consequently, CMsiRNAdb exhibits a significant biological bias toward specific cellular lines and metabolic target genes, restricting its utility for training robust efficiency-predictive models [12].

To bypass this scalability bottleneck, Large Language Models (LLMs) have emerged as powerful tools for mining complex and unstructured literature. They have been successfully deployed across various scientific domains to automate data extraction. Examples range from extracting synthesis conditions from tables in materials science to mining molecular structures directly from chemical patents [31, 34]. Applying similar LLM-driven approaches to siRNA patent mining could resolve current structural barriers and enable the curation of larger datasets. This LLM-driven approach was recently validated in the field of nucleic acid therapeutics through the development of the ASO Atlas. By leveraging advanced LLMs to process unstructured XML patent tables and standardize chemical annotations, researchers successfully extracted over 188,000 antisense oligonucleotide (ASO) sequences from 417 U.S. patents with minimal transcription errors [13]. Implementing a similar LLM-based parsing pipeline for siRNA presents a significant opportunity. By automating the extraction of sequence configurations, structural modifications, and *in vitro* efficacy metrics across hundreds of unmined patents, we can expand the current data landscape, mitigate existing biases, and build a truly comprehensive training dataset to create state-of-the-art deep learning models.

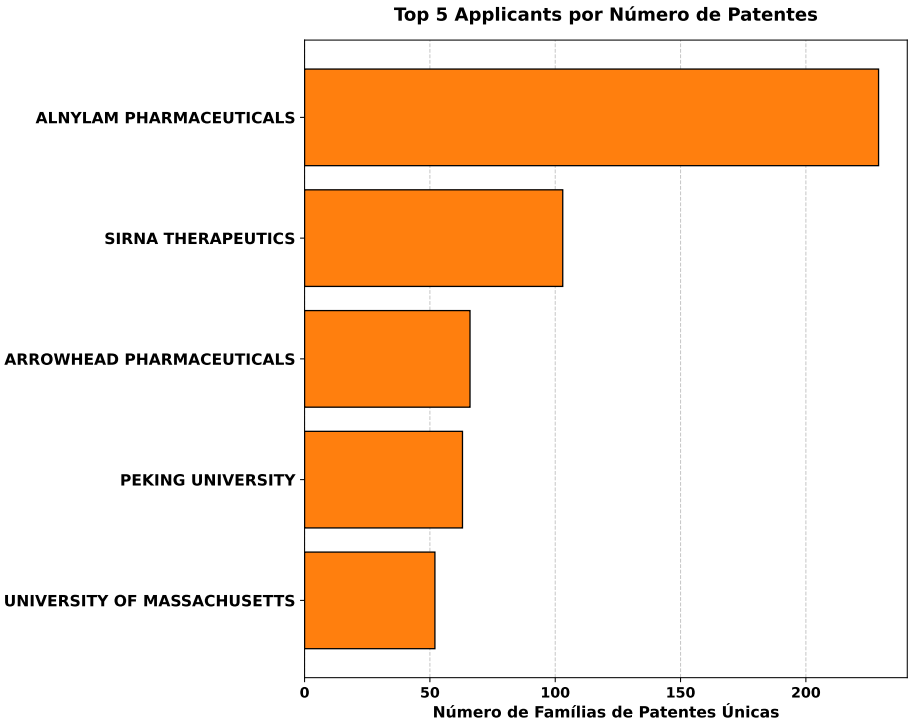

Fig. 2. Top 5 Applicants by Patent Families in the siRNA Technology Landscape. The chart illustrates the number of unique patent families held by various academic and corporate organizations. Patent metadata was retrieved via the EPO API and consolidated using an automated entity resolution pipeline.

2 Project Objectives

The goal of this project is to develop a reusable, LLM-driven pipeline that converts unstructured patent literature into Machine Learning (ML)-ready datasets containing siRNA sequences, chemical modifications, and experimental metadata. Using Alnylam Pharmaceuticals as a proof-of-concept case study, the project aims to:

- **Pipeline Development:** Design a scalable, LLM-based workflow to process unstructured text within patent documents retrieved via the EPO Open Patent Services (OPS) API.
- **Data Curation:** Apply the pipeline to the Alnylam Pharmaceuticals patent portfolio to curate an dataset mapping modification patterns within siRNA sequences to specific experimental contexts (cell lines, targets, and assays).
- **Statistical Validation & Quality Assessment:** Perform comparative analyses to identify efficacy trends and validate the biological relevance of the automatically mined data.

3 Proposed Methodology

To achieve these objectives, the project will focus first on robust data engineering followed by thorough statistical validation to assess dataset quality.

3.1 Data Processing and LLM-Driven Extraction

The first stage utilizes a custom Python-based API client to retrieve JSON-formatted data from the European Patent Office (EPO) database, filtering exclusively for applications associated with Alnylam Pharmaceuticals. To transform these files into a structured format, an LLM-driven extraction will be used to obtain the following data:

- **Molecular Data:** Sense and antisense sequences, chemical modification patterns (for example, 2'-OMe, 2'-F), and the targeted genes.
- **Experimental Metadata:** SiRNA concentrations, cell lines, and transfection methods.
- **Efficacy Metrics:** Quantitative silencing results (for example, % mRNA knockdown) and IC_{50} values.

Afterward, a Python script performs an automated data integrity check, ensuring all extracted sequences and chemical notations adhere to biological rules.

3.2 Statistical Analysis and Biological Coherence

Following dataset construction, we will use Python libraries (Pandas, SciPy, Seaborn) to perform statistical analysis. The goal is to validate the biological relevance by identifying if the extracted trends align with established pharmacological patterns. This includes:

- Distribution analysis of silencing efficacy across different chemical modification types.
- Correlation matrices to identify relationships between specific chemical patterns and biological activity.

4 Work Plan

The tasks of the project are organized as follows:

- **Data Retrieval & LLM Engineering (April 13 - May 3):** Extraction of patent data via OPS API, prompt engineering for LLM parsing, and initial data structuring.
- **Data Cleaning & Statistical Validation (May 4 - May 17):** Automated integrity checks, biological coherence validation, and performing distribution and correlation analyses.
- **Manuscript Drafting & Presentation Prep (May 18 - June 4):** Drafting the full manuscript and preparing for the presentation on May 27.
- **Supervisor Review (June 5 - June 18):** Two-week window for feedback, final corrections, and polishing the 12-page article.
- **Final Submission (June 19):** Final submission of the article.

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